

MOTOR VEHICLE COLLISION

Guided by

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By

Group – 12

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ABSTRACT

The result in a sizable number of injuries, fatalities, and economic losses globally, motor vehicle accidents (MVCs) are a serious public health problem. This summary gives readers a thorough review of MVCs, looking at their frequency, causes, and effects. The study covers the importance of environmental elements, such as weather and road conditions, as well as human factors, such as driver attention, impairment, and exhaustion, in the occurrence of MVC. It also looks at the role that technology, safety features, and vehicle design have in reducing the severity of crashes. Furthermore, to lessen the incidence of MVCs, the abstract explores the significance of efficient road traffic legislation, enforcement, and education. This abstract provides insightful information with an emphasis on preventative measures and injury prevention methods.

Data Source:

The dataset is open-source, and it is available to the public on the website Kaggle.

Source Link: <https://www.kaggle.com/datasets/utkarshx27/motor-vehicle-collisions>

Data Description:

This dataset contains 1987321 measurements gathered between 2012 and 2023.

Variable	Description
Data & Time	Data and time have been recorded
Zip-code, street name, Borough	Specify location by the zip-code, street name, borough of the city
Locations (Latitude, Longitude)	Gives the exact location where the accidents have happened with coordinates
Contributing factors	Cause of the accidents and reasons
Injured/killed-Pedestrians, Persons, Cyclist, Motorist	The place where the accident took place
Vehicle Type	Vehicle types involved in accident

Data Understanding:

The shape of Dataset: 1987321

Sample of Data

After removing all the null values, The shape of Dataset: 1319415

```
]: print(newdf3)
```

The following order is being followed to achieve are result:

1. Loading Data
2. Exploratory Analysis
3. Cleaning data
4. Merging weather data from External source & Extrapolation of missing data
5. Data Visualization & Insights

Data Visualization:

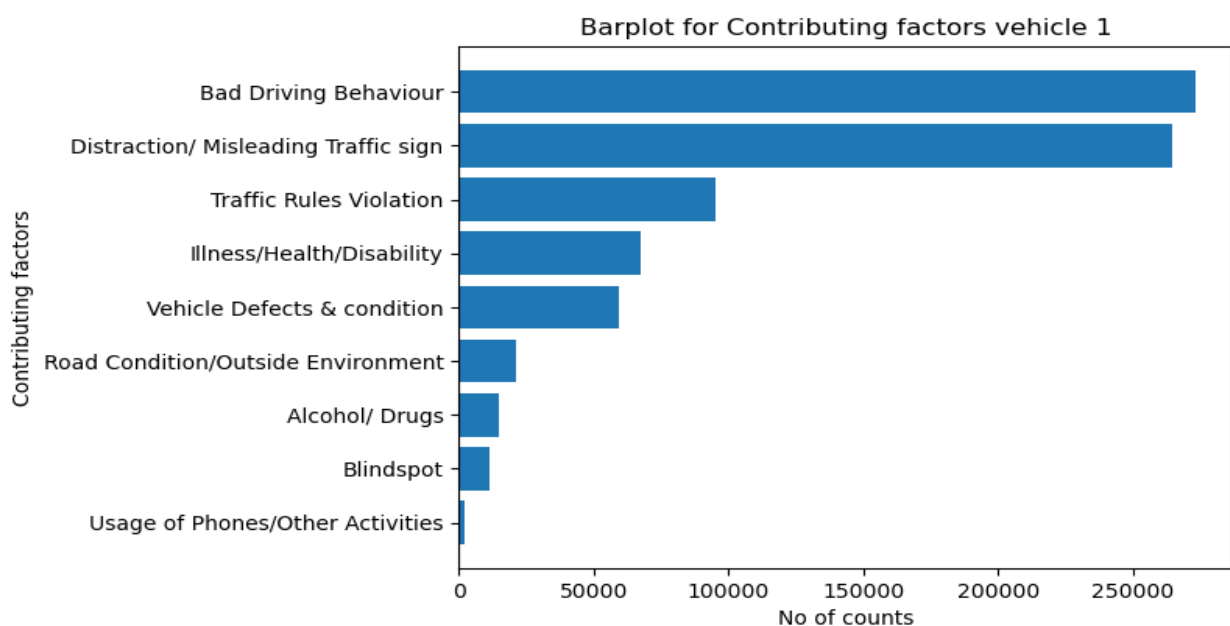


The above Graph gives us all the Contributing factors for why the accidents are happening.

Now we have combined the duplicate and similar into a few categories.

- Unspecified
- Bad Driving Behavior
- Distraction/ Misleading Traffic sign
- Traffic Rules Violation
- Illness/Health/Disability
- Vehicle Defects & condition
- Road Condition/Outside Environment
- Alcohol/ Drugs
- Blindspot
- Usage of Phones/Other Activities

INSIGHT: 1 Finding the category in the column Contributing factor vehicle 1



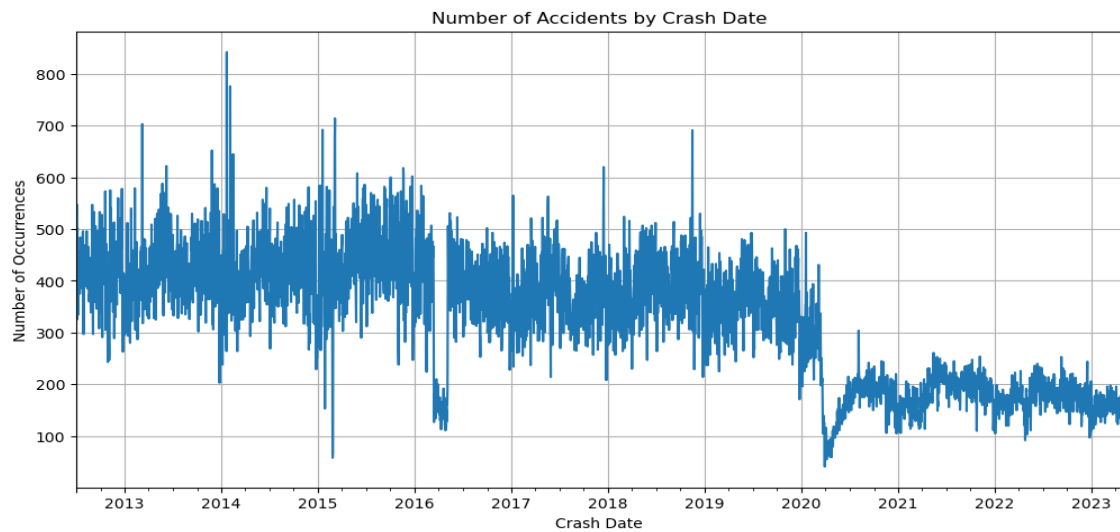
INFERENCE:

The data depicted in the graph highlights that the highest number of accidents stem from instances of Bad Driving Behavior, encompassing actions such as Tailgating, Reckless Driving, Unsafe Passing, Inexperienced Driving, Hazardous Lane Changes, Unwarranted Responses to Unrelated Vehicles, Aggressive Driving or Road Rage, Unjustified Reactions to Other Uninvolved Vehicles, Vehicle Vandalism, Improper Passing or Lane Usage, Incorrect Turns, and Failure to Yield Right-of-Way.

Additionally, Accidents attributed to Distractions and Misleading Traffic Signals also prominently contribute to the overall accident count. This category encompasses External Distractions, Inadequate

or Misleading Lane Markings, Use of On-Board Navigation Devices, Malfunctioning Traffic Control Devices, and Driver Inattention or Distraction.

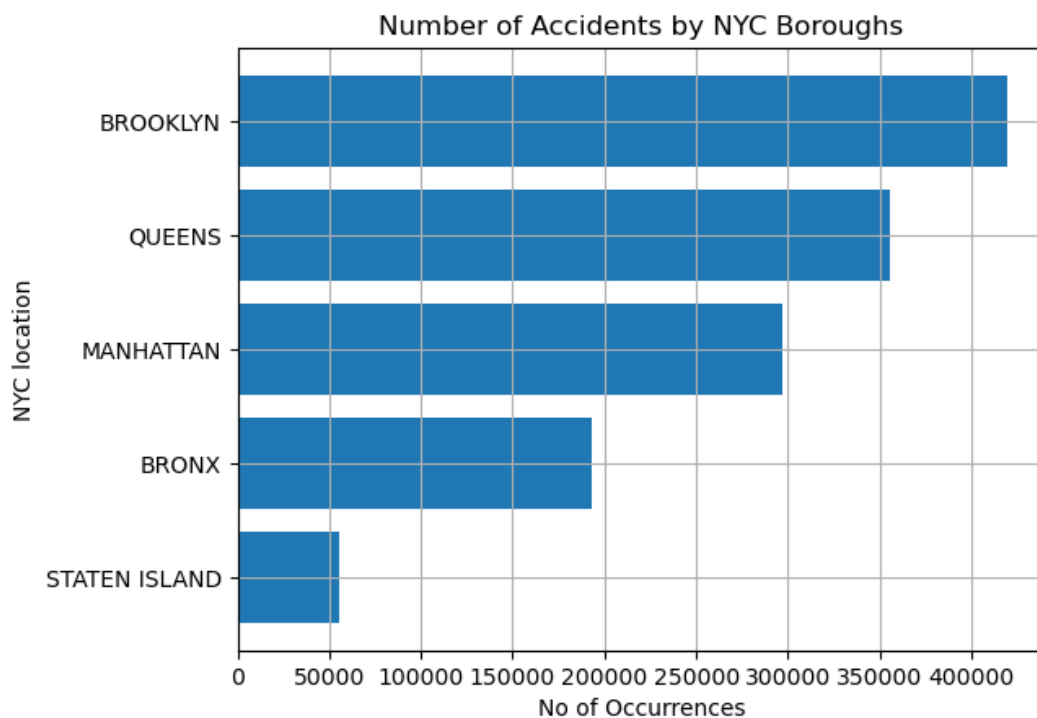
INSIGHT: 2 Number of accidents date wise



INFERENCE:

A clear correlation exists between road traffic conditions and the frequency of accident occurrences. Notably, there is a sharp decline in accidents during the initial two quarters of 2020, corresponding to the period encompassing and following the lockdown measures.

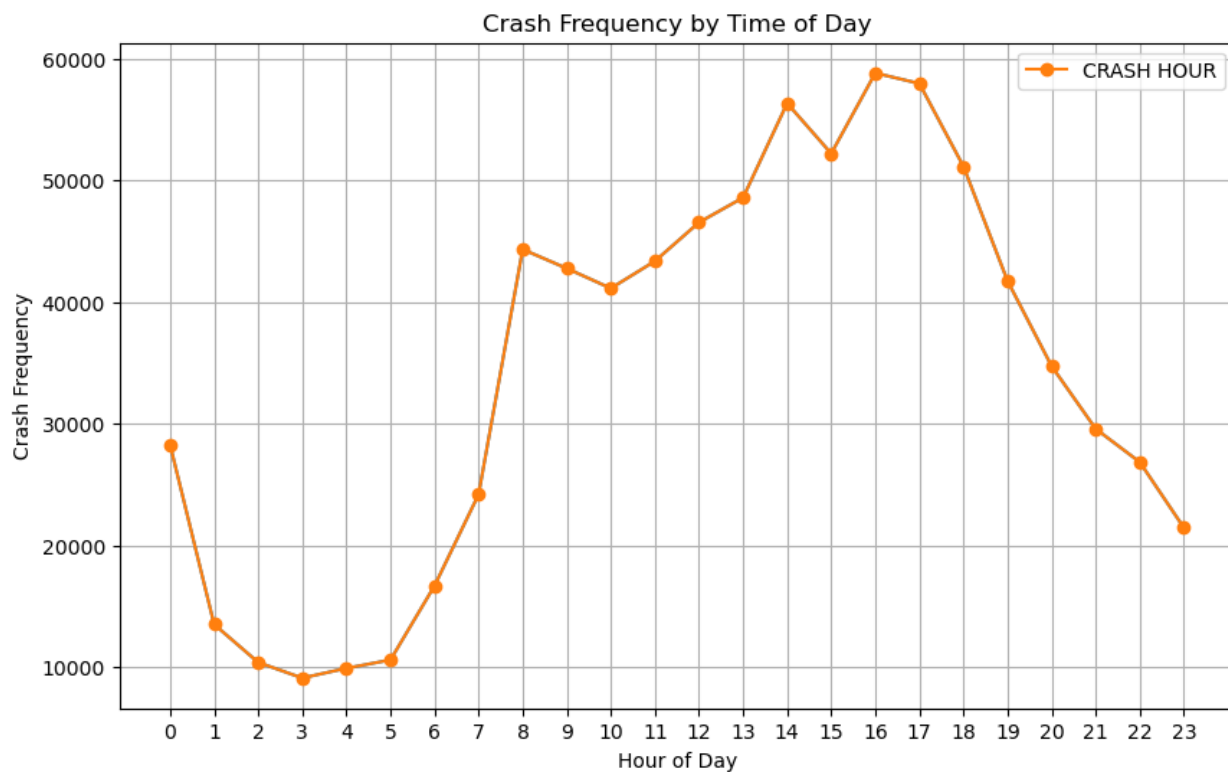
INSIGHT: 3 Number of accidents based on the BOROUGH



INFERENCE:

Brooklyn surpasses both Queens and Manhattan in terms of accident count, which can be attributed to its larger population size compared to other boroughs, establishing a direct correlation.

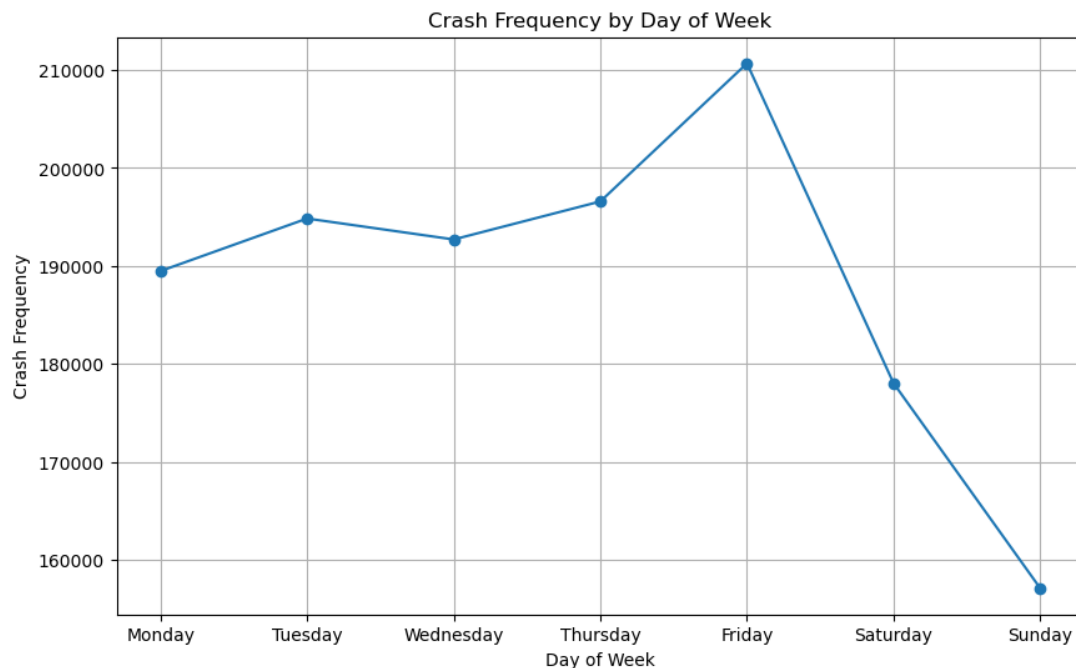
INSIGHT: 4 Scatter plot for the time zones of the data



INFERENCE:

It can be deduced that the period between 3 pm and 6 pm consistently records the highest number of accidents, indicating heightened activity during these hours. Similarly, there is a noticeable surge in accident occurrences between 7 am and 8 am, which coincides with the commencement of office and school hours, demonstrating a distinct pattern aligned with the start and end of these activities.

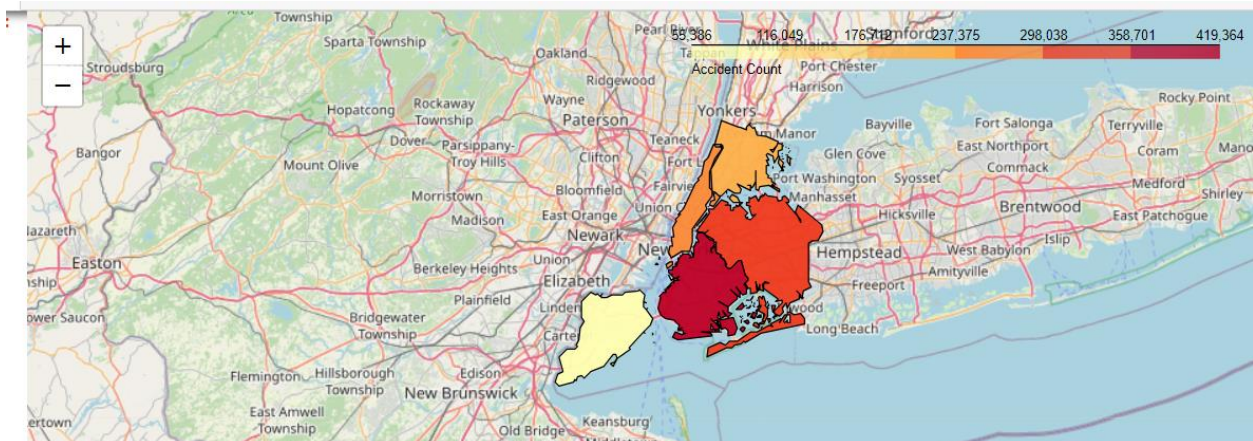
INSIGHT: 5 Crash frequency based on day of the week (Sunday to Saturday)



INFERENCE:

Particularly on Fridays, the highest number of accidents have been documented, suggesting a greater likelihood of accident incidents compared to other days of the week. This count is approximately 22,000 accidents above the recorded average. In contrast, the lowest number of accidents is observed on Sundays.

INSIGHT: 6 Plotting spatial pattern heat map of accidents by (latitude, longitude) data

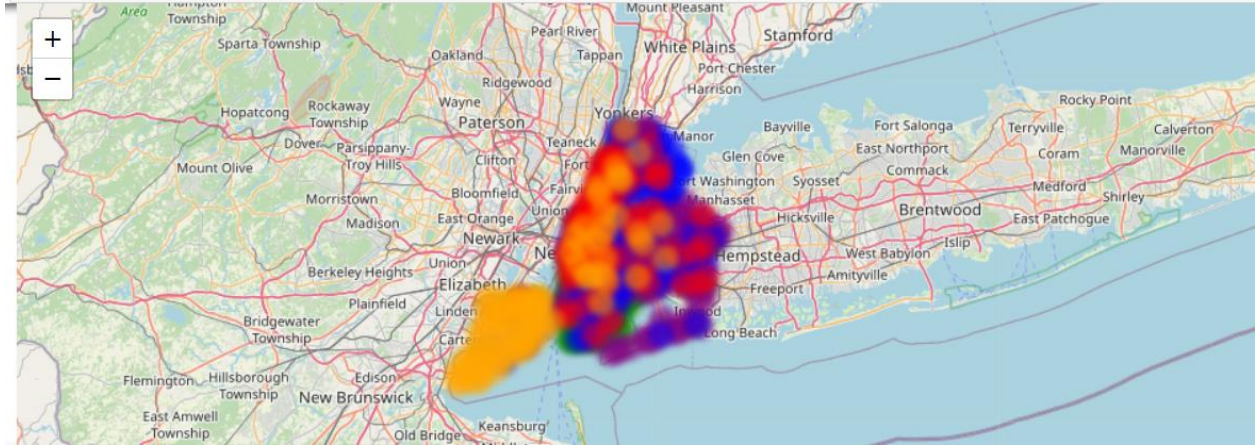


INFERENCE:

Analyzing the heat map of accidents, a discernible pattern emerges wherein accidents are prevalent at all signalized intersections and road junctions. Notably, there is no instance of a signalized location without

recorded accidents. The concentration of incidents is notably elevated near hospitals, pharmacies, as well as landmarks and monuments.

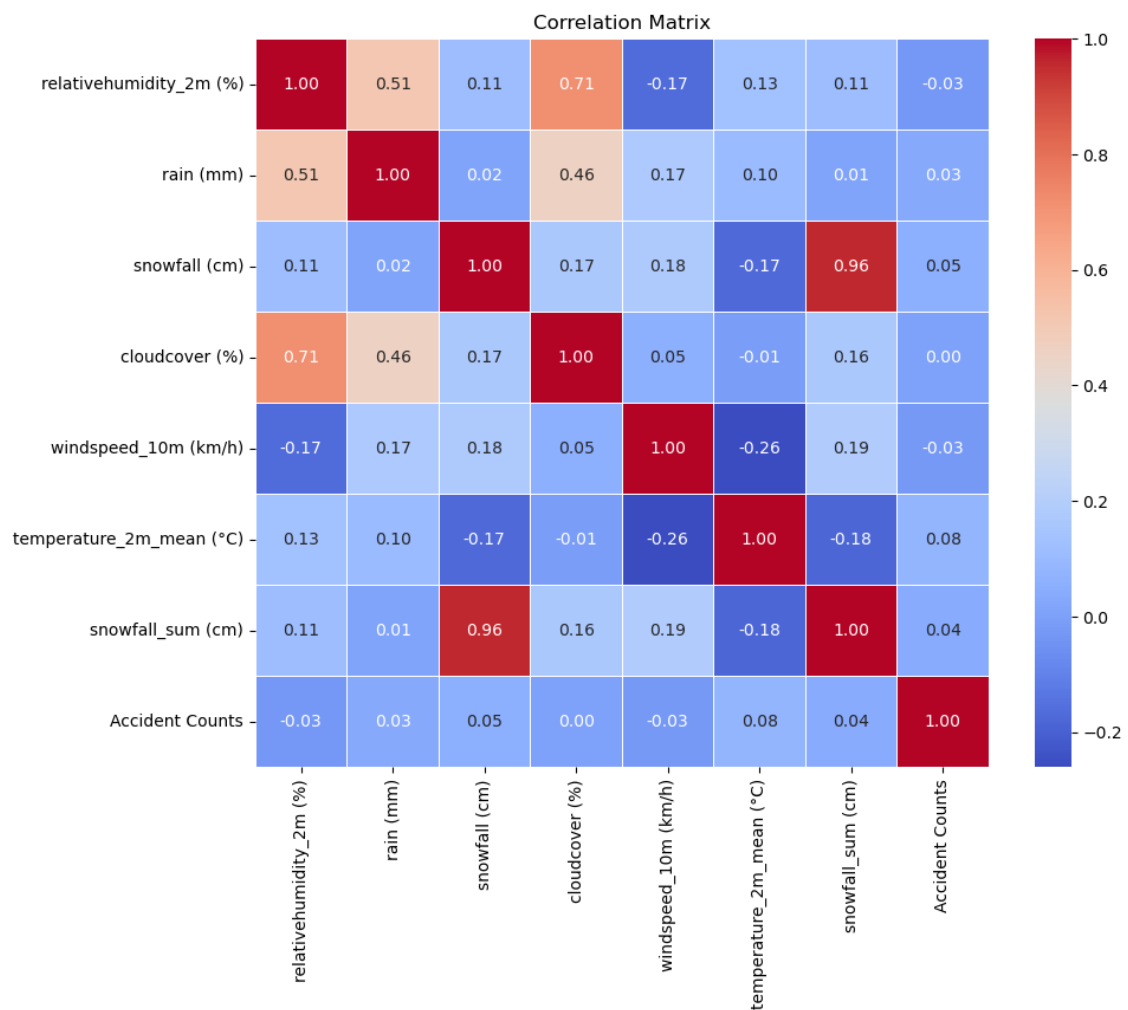
INSIGHT: Based on the heat map of the accidents it can be inferred that almost at all the signals and intersection of the roads there was an accident recorded



INFERENCE:

Apart from the previously highlighted regions, several other areas exhibit a notable frequency of accidents. The Bronx, for instance, experiences a considerable number of accidents, due to its dense population and heavy traffic flow. Similarly, Queens, being a vast borough, also witnesses a substantial volume of accidents, attributed to its extensive road network and high traffic levels. It is worth noting that hospitals and monuments contribute to the occurrence of accidents, particularly along highways, with a pattern spanning from Manhattan to Staten Island, then to the Bronx, and to Queens.

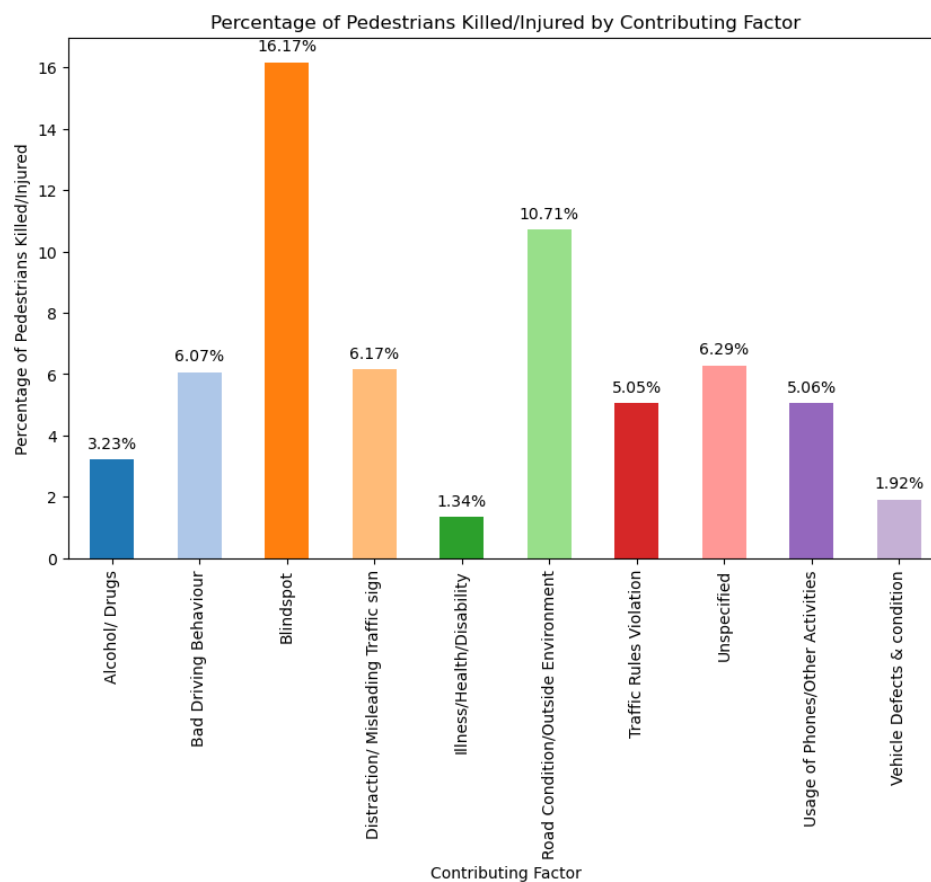
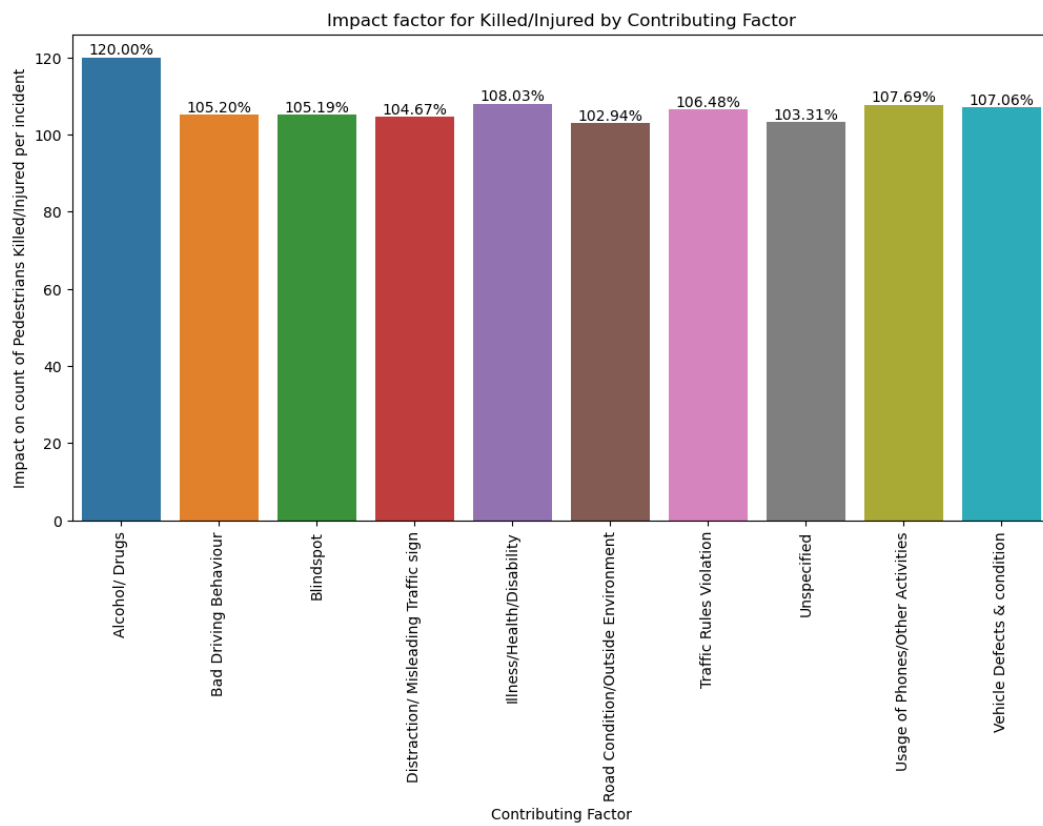
INSIGHT: 7 Correlation Matrix

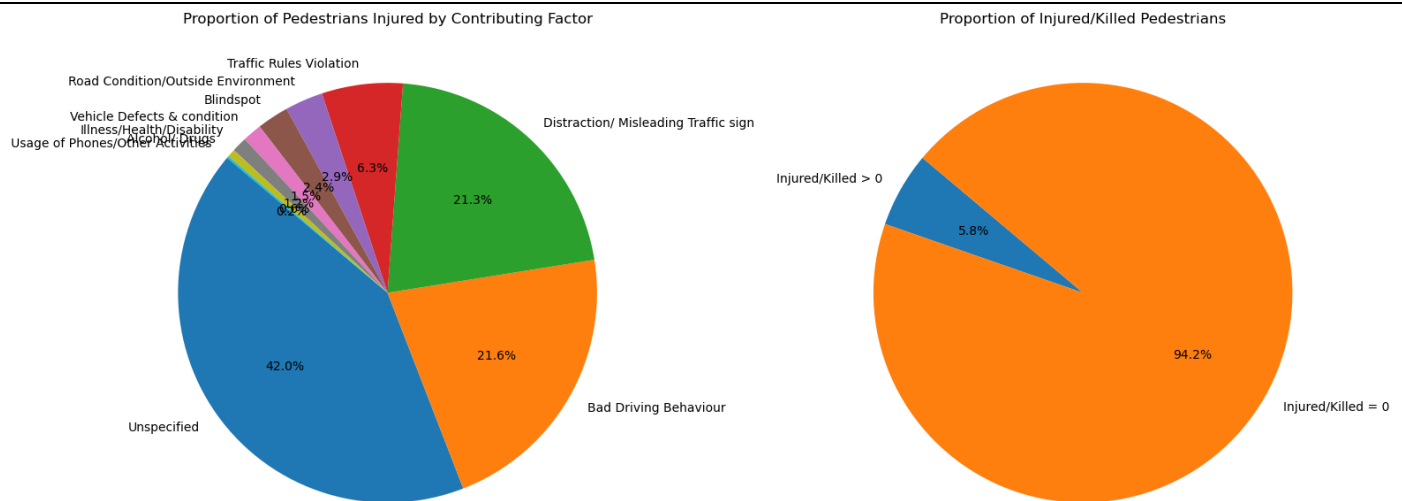


INFERENCE:

It is evident that there is no direct correlation between weather conditions and the number of accidents on any given day.

INSIGHT: 8 Percentage of Pedestrian killed/injured by Contributing factor

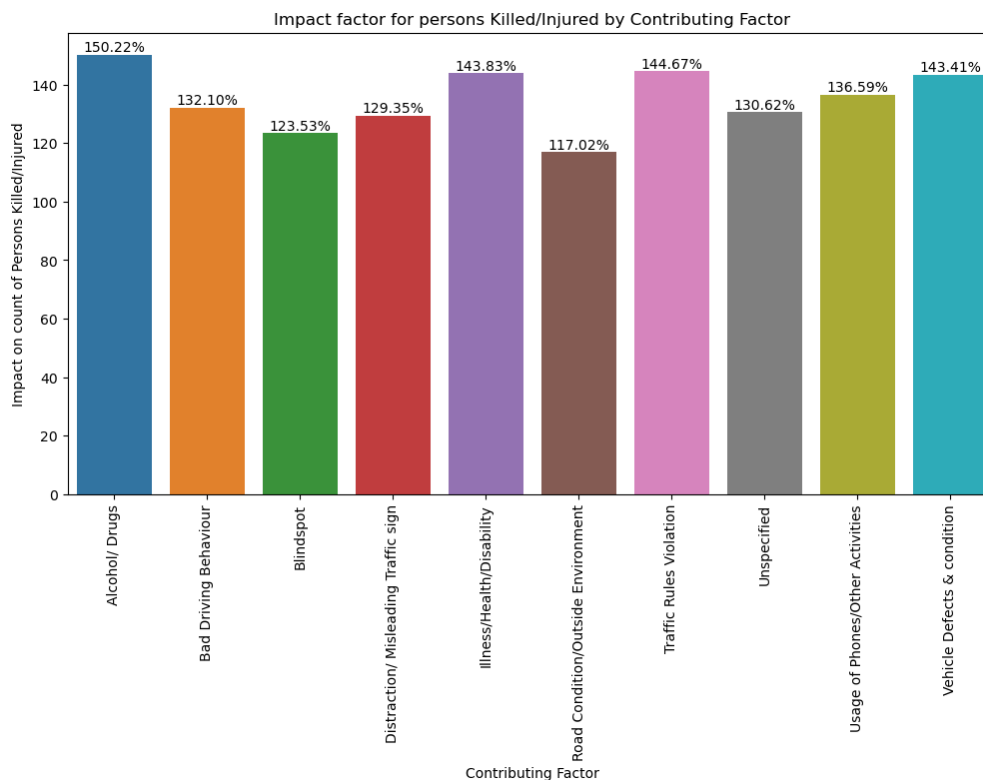


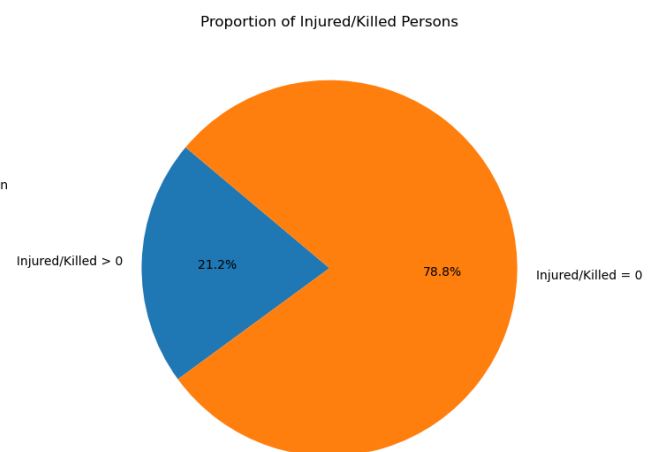
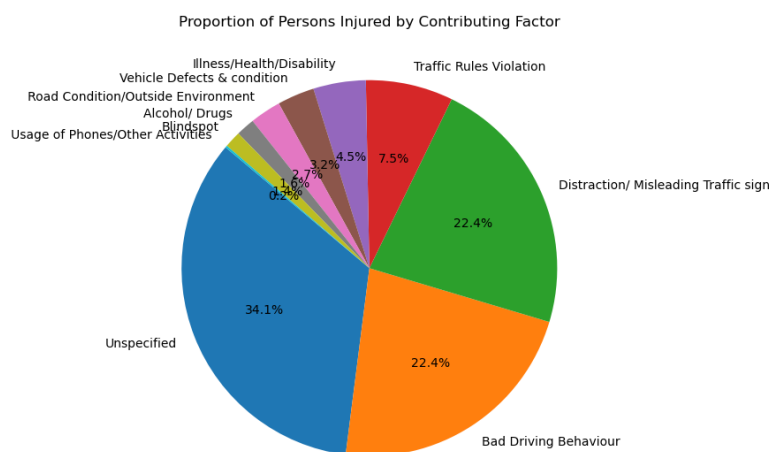
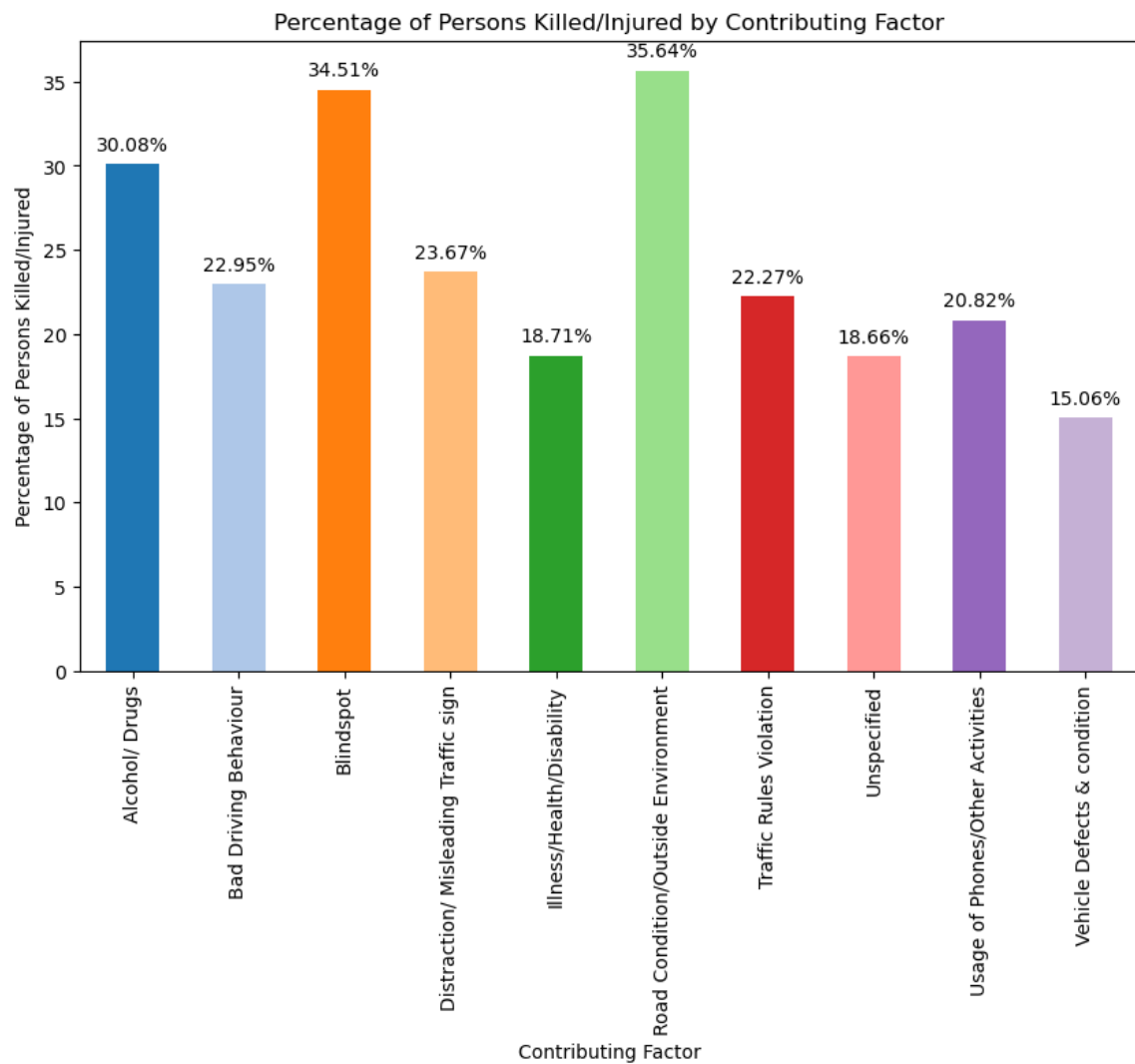


INFERENCE:

- Among no of pedestrians killed/injured the most impactful factor is Alcohol Drugs means the number of people killed or injured per incident is more than other factors
- Among the total pedestrians killed or injured the major factor is contributed by Blindspot Scenarios
- Percentage of pedestrians killed or injured by Contributing factor

INSIGHT: 9 Percentage of Persons Killed or Injured by Contributing factor

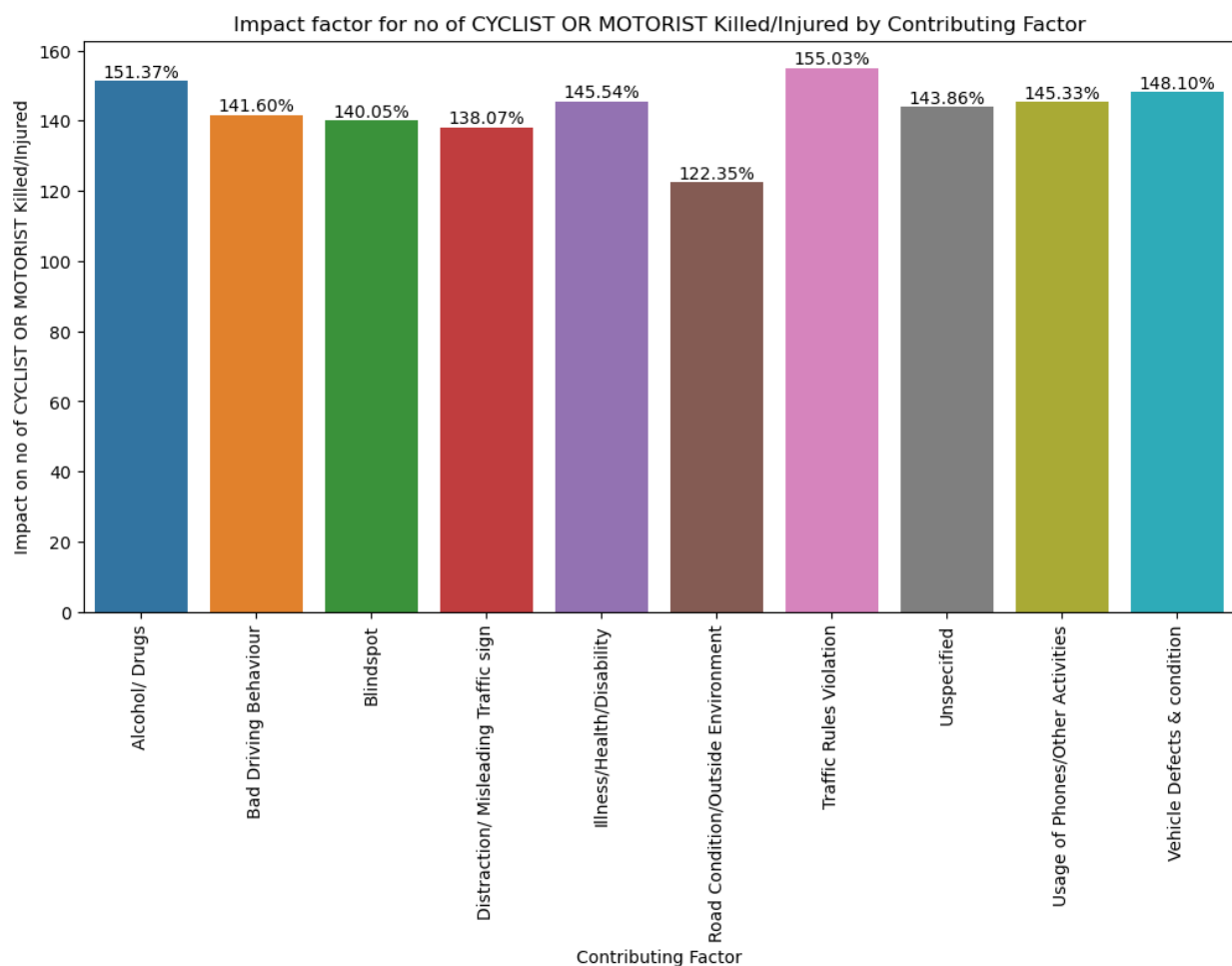


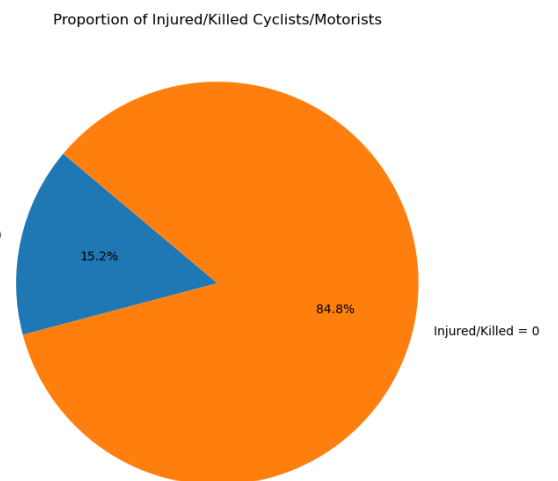
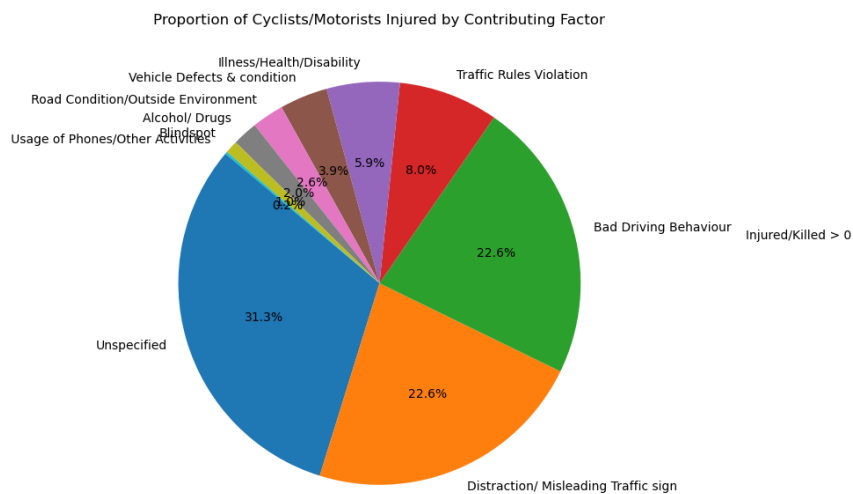
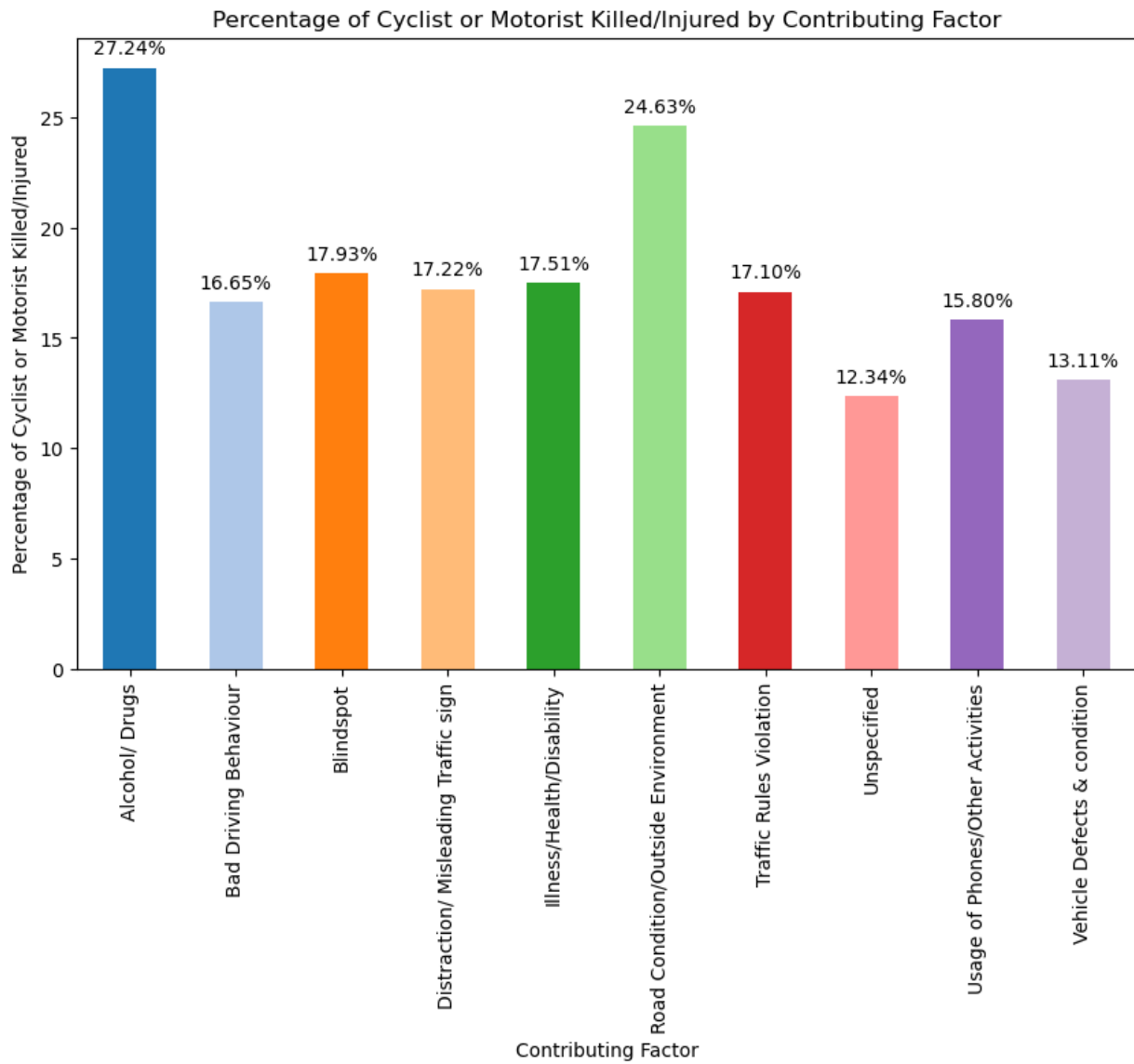


INFERENCE:

- i) Among no of persons killed/injured the most impactful factor is Alcohol Drugs means the no of people killed or injured per incident is more than other factors
- ii) Among the total persons killed or injured the major factor is contributed by Road Condition or Outside Environment Scenarios
- iii) Percentage of persons killed or injured by Contributing factor

INSIGHT: 10 Percentage of Cyclist/Motorist killed or injured by Contributing factor





INFERENCE:

- i) Among no of Cyclist/Motorist killed/injured the most impactful factor is Traffic rules violation means the number of people killed or injured per incident is more than other factors
- ii) Among the total Cyclist/Motorist killed or injured the major factor is contributed by Alcohol/Drugs Scenarios
- iii) Percentage of Cyclist/Motorist killed or injured by Contributing factor

Challenges Faced:

- ❖ **Data Quality and Cleaning:** Dealing with a dataset of such magnitude (over 1.9 million records) can be challenging, especially when it contains missing or erroneous values. Cleaning the data and managing null values require careful consideration and may involve making assumptions or using imputation techniques. We have used CSVviewer software to explore the dataset csv file completely to sort and view.
- ❖ **Data Extrapolation:** The missing values like Zip-code, Borough details were extrapolated using the coordinates values which took lot of time but due to system constraint and technical limitation we could not able to apply it fully to the dataset but a trial for 500 records were done.
- ❖ **Data Merging:** Weather data were not available in the original dataset so external source data is found matching exact timeline and it was merged accordingly to right primary key which is Crash date and time.
- ❖ **Categorizing Causes of Accidents:** Combining duplicate and similar accident causes into meaningful categories can be subjective and might require domain expertise. There could be ambiguity in certain cases, and deciding how to group them can impact the analysis results
- ❖ **Managing External Factors:** Analyzing motor vehicle collisions often involves considering external factors like weather, road conditions, and traffic density. These factors can be dynamic and hard to control for, making it challenging to isolate their specific impact on accident occurrences.
- ❖ **Data Visualization:** Creating meaningful and insightful visualizations from a large dataset can be difficult. Choosing appropriate visualization techniques and presenting the data in a way that effectively communicates the information can be challenging.
- ❖ **Interpretation of Insights:** Drawing accurate and actionable insights from data analysis is crucial. However, interpreting the results correctly and understanding the implications of the findings requires careful attention and domain knowledge.
- ❖ **Generalization of Findings:** While the dataset used for analysis may be substantial, the findings might not be representative of the entire population of motor vehicle collisions. Generalizing the results to a broader context needs to be done with caution.
- ❖ **Data Source Limitations:** Depending on the dataset's source, there may be potential biases or limitations. Understanding the data collection methodology, potential sampling biases, and coverage issues is essential for accurate interpretation.

Conclusion:

In conclusion, this study provides valuable insights into the prevalence, causes, and consequences of Motor Vehicle Collisions (MVCs). The dataset analysis, covering a vast range of accidents from 2012 to 2023, highlights the significance of addressing road safety as a critical public health concern. The findings shed light on numerous factors contributing to MVC occurrences, ranging from environmental elements such as weather and road conditions to human factors like driver behavior, distraction, and impairment.

The analysis emphasizes the role of technology, safety features, and vehicle design in mitigating the severity of accidents. These findings underscore the importance of continuous advancements in vehicle safety technology and the need for stricter safety regulations and enforcement to prevent MVCs.

The presented insights reveal critical patterns and trends in MVC occurrences, such as higher accident counts during specific hours of the day and weekdays. Additionally, certain geographic areas, particularly those with high population density, show a higher incidence of accidents, suggesting the importance of targeted safety measures in these regions.

Based on the analysis, injury prevention strategies should focus on addressing bad driving behavior, distractions, and misleading traffic signs. These findings underscore the need for public awareness campaigns to educate drivers and pedestrians about road safety and responsible behavior.

Furthermore, the study highlights the significance of preventive measures, injury prevention methods, and continuous monitoring of accident data to identify emerging patterns and adapt safety measures accordingly.

The insights provided by this analysis serve as a foundation for policymakers, road safety experts, and stakeholders to develop evidence-based interventions and policies aimed at reducing motor vehicle collisions, minimizing injuries, and saving lives. By incorporating technology, improving infrastructure, and promoting responsible driving habits, we can work towards creating safer roads for all and achieving considerable progress in reducing the toll of motor vehicle accidents on society.